MILITARY SPECIFICATION

SWITCHBOARD, AUDIO, DESIGN, TEST AND MANUFACTURE OF

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the design, test and manufacture of an audio switchboard. This unit shall be capable of transferring audio signals from a receiver to one or more remote control stations on all naval (shipboard) platforms.

1.2 Classification. The audio switchboard shall be type SB-2727/SRR, or equivalent.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-16400	Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore, General Specification for
MIL-S-19500	Semiconductor Device, General Specification for

STANDARDS

MILITARY

MIL-STD-108	Definition of and Basic Requirements for Enclosure for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-189	Rack, Electrical Equipment, 19 Inch and Associated Panels
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by contracting officer.)

3. REQUIREMENTS

3.1 General. The switchboard shall be in accordance with MIL-E-16400 except as hereinafter specified.

3.1.1 Standard electronic modules. The use of standard electronic modules is not applicable to this specification.

3.1.1.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.2).

3.2 Performance requirements. The switchboard shall be capable of transferring the audio output from a radio receiver to a varying number of remote control stations.

3.2.1 Matrix size. The switchboard shall be designed in a matrix of twenty-four (inputs) by twenty (outputs). Each input and output shall be a two-wire configuration. Any single input shall have the capability of being transferred from one to twenty outputs.

3.2.2 Signaling handling capability.

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3.2.2.1 Signal frequency. The switchboard shall have the capability to transfer signals ranging from direct current (DC) to 3500 Hertz (Hz).

3.2.2.2 Signal levels. The switchboard shall be capable of transferring signal levels ranging between -10 dBm to + 4dBm.

3.2.2.3 Impedance. The switchboard shall be capable of successfully operating in circuits where the source and load impedances are 600 ohms, balanced or unbalanced.

3.2.3 Capacity. Each rotary switch shall be capable of transferring a maximum of 125 ma at a voltage of 125 volts, Root Mean Square (RMS) or DC.

3.2.4 Insertion loss. The insertion loss of any individual circuit shall be 0 dB ± 1dB. A circuit shall be defined as a connection of any input to any output.

3.2.5 Isolation. The isolation between any set of input pairs, or any input and output pairs, or any output pairs shall be no less than 60 dB.

3.2.6 High potential. The switch shall be capable of withstanding a high potential voltage of 500 volts, RMS or DC, between any mutually insulated points and from any point to ground.

3.3 Physical requirements.

3.3.1 Enclosure. The switchboard shall be dripproof (45 degrees) in accordance with MIL-STD-108.

3.3.2 Weight. The switchboard shall be designed not to exceed 20.412 kilograms (forty-five pounds).

3.3.3 Size. The switchboard, in accordance with MIL-STD-189 rack size A, shall be designed for standard 482.60 millimeters (19 inch) rack and shelf mounting. The size of the switchboard shall not exceed 457.20 millimeters (18 inches) in height, nor 304.80 millimeters (12 inches) in depth.

3.3.4 Interchangeability. The switchboard shall be a functional replacement for the currently used switchboard, SB-2727/SRR.

3.3.5 Cable entrance. Flush type removable entrance plates shall be located on the top, bottom, and rear of the enclosure to facilitate installation of entrance bushings for external connecting cables.

3.4 Environmental conditions. The switchboard shall conform to the environment requirements specified herein.

3.4.1 Vibration. The switchboard shall conform to the vibration requirements of MIL-E-16400 for shipboard equipment.

3.4.2 Shock. The switchboard shall conform to the shock requirements of MIL-E-16400 for shipboard equipment. The switchboard shall conform to all performance requirements after shock.

3.4.3 Electromagnetic. The switchboard shall conform to the requirements of MIL-STD-461, Class IC, except the limit for test requirement CS06 shall be 400 volts peak and 5.0 microseconds wide.

3.4.4 Operating/non operating temperature range. The switchboard shall conform to all requirements of MIL-E-16400, Range 4.

3.4.5 Salt fog (spray). The switchboard shall conform to all requirements of MIL-E-16400.

3.5 Reliability and maintainability requirements.

3.5.1 Switch life. Rotary switch types must have a minimum switch life of 25,000 cycles, where a cycle is defined as one complete rotation (360 degrees) of a rotary switch knob.

3.5.2 Quantitative maintainability requirements. The switchboard shall have a specified median corrective maintenance time (Mct) of 0.25 hours and a maximum corrective maintenance time (Max ct) of 0.75 hours at a 90 percent confidence level when repair is accomplished at the organizational level of maintenance by the replacement of lowest sub-assemblies and chassis-mounted components

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within [illegible] calendar days. IQ Bid or per Federal agency grant of

(this includes electronic, electrical, electro-mechanical and mechanical parts). The equipment Mct includes localization, isolation, disassembly, interchange, reassembly, alignment and check out for all maintenance tasks.

3.6 Maintenance design. Parts shall be easily replaceable from the front of the equipment without special tools or skills and with minimum disassembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or the purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor shall be subject to Government verification at any time. Verification shall consist of (a) surveillance of the operations to determine the practices, methods and procedures of the written quality program are being properly applied, (b) Government product inspection to measure quality of product inspection of delivered products to assure compliance with all inspection requirements of the specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or when conformance of product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-108.

4.2 Classification of inspection. The method of examination and testing of equipment falls within the following classifications:

(a) First Article Inspection (see 4.3)
(b) Quality Conformance Inspection
    (1) Production Inspection (see 4.4.1)
    (2) Environmental Inspection (see 4.4.3)
(c) Reliability Testing (see 4.6)
(d) Maintainability Demonstration (see 4.7)
(e) Inspection of Preparation for Delivery (see 4.8)

4.3 First article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the test specified in TABLE I.

TABLE I. First article inspection.

Tests and Inspections	Requirement paragraph	Method paragraph
Surface Examination	3.1, 3.3.1, 3.3.2, 3.3.3	4.5.1
Operating Test	3.2	4.5.2
Weight & Dimensions	3.3.2, 3.3.3	4.5.1
Enclosure Test	3.3.1	4.5.1
Temperature Test	3.4.4	4.5.1
Switch Life Test	3.5.1	4.6
Electromagnetic Interference	3.4.3	4.5.3
Shock, Vibration, Salt Spray	3.4.2, 3.4.1, 3.4.5	4.5.1

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4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests, which detect deviations from design, tests of adjustment and tests which detect hidden defects of material. Production inspection shall include the tests and examinations of TABLE II.

TABLE II. Production inspection.

Inspection	Requirement paragraph	Method paragraph
Surface Examination	3.3.1, 3.3.2, 3.3.3	4.5.1
Operating Test	3.2	4.5.1

4.4.2 Environmental inspection. Environmental inspections shall be as listed in TABLE III and shall be performed on units which have been subjected to and passed production inspection.

TABLE III. Environmental inspection.

Inspection	Requirement paragraph	Method paragraph
Temperature Test	3.4.4	4.5.1
Shock, Vibration	3.4.1, 3.4.2	4.5.1
Salt Spray Test	3.4.6	4.5.1
Enclosure	3.3.2	4.5.1

4.4.2.1 Environmental inspection samples. Samples for the environmental inspection shall be selected in accordance with the following:

(a) One sample from the first 25 production units fabricated.
(b) One sample from each successive 75 units or fraction thereof, procured.

4.4.3 Non-conformance and retest. If a sample unit fails an inspection specified in 4.4.2, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes, and so forth. The equipment shall then be retested.

4.4.4 Corrective action. If corrective action results in a change of a part or parts, electrical or mechanical layout or function, or if the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred immediately to the contracting officer. It shall then be determined by the Command or Agency concerned whether correction should be made to the equipment already delivered under the contract.

4.4.5 Reinspection of conforming production control and environmental sample units. Unless otherwise specified, sample units which have been subjected to and passed production control or environmental inspections, or both, may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage.

4.5 Inspection procedures.

4.5.1 Tests and examinations. Test and examinations shall be conducted in accordance with the applicable inspection procedures and test methods of MIL-E-16400, unless otherwise specified herein.

4.5.2 Operating test. The equipment shall be energized and subjected to an operating test to demonstrate proper functioning at a sufficient number of points within the specified range to determine compliance with all the performance requirements of 3.2.

4.5.2.1 Insertion and isolation test. Equipment shall be checked for insertion loss (3.2.4) and isolation (3.2.5) utilizing the standard test tone frequency of 1000 ±25 Hz introduced at an input level of -4 dBm.

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4.5.3 Electromagnetic compatibility (EMC). The equipment shall conform to the requirements specified in 3.4.3 when tested in accordance with MIL-STD-462.

4.6 Reliability testing.

4.6.1 Rotary switch life. Each rotary switch type shall be subjected to a contact reliability test to demonstrate a minimum life of 25,000 operations before failure. For purposes of this test, switches shall be considered of the same type when they are manufactured by the same contractor, to the same physical specifications, using identical materials and assembly processes. The switch life test shall be conducted prior to completion of first article inspections.

4.6.2 Test conditions. Testing shall be performed at an ambient temperature of ± 5OC. Input current and voltage levels shall be as specified in 3.2.3. The switching rate shall be determined by the contractor but shall not result in excessive internal heating of the device. A switching sequence shall be established which will approximate actual service usage of the switch when installed in the equipment.

4.6.3 Accept/reject criteria. Accept/reject criteria shall be in accordance with TABLE C-1 of MIL-S-19500 using a lot tolerance percent defective of 20 and an acceptance number (C) not to exceed 2. Each test sample shall be cycled for a minimum of 25,000 cycles and shall be considered defective if it experiences a contact failure or exhibits intermittent electrical operation prior to 25,000 cycles.

4.6.4 Disposition of test samples. Switches subjected to the life test shall neither be installed in deliverable equipments nor delivered to the Government as part of a contract quantity of spare parts.

4.7 Maintainability demonstration. The contractor shall perform a maintainability demonstration at the organizational level. The organizational level demonstration shall be performed by a technician with the experience equivalent to that of a Petty Officer, third class. The demonstration shall be used to verify conformance to the equipment Mct. requirements of the contract.

4.7.1 Demonstration conditions. The equipment maintainability requirements shall be demonstrated in accordance with Method 10, MIL-STD-471 (except that the sample size shall be 20 faults) by replacement of subassemblies and chassis-mounted electronic, electrical, electro-mechanical, and mechanical components or parts at the organizational level. The contractor shall determine 20 candidate faults for this demonstration. For fault simulation, a number shall be assigned to each component/part within a subassembly or chassis-mounted component/part. Fault simulation for modules shall be accomplished by introduction of faulty parts corresponding to the number selected.

TABLE IV. Acceptance table.

Acceptance	Sample Size	Confidence level percent	Acceptance level percent
Mct	20	90	0
Mmaxct	20	90	0

4.7.2 Technicians. The Command or Agency concerned reserves the right to provide and select the technicians to perform the equipment maintainability demonstration.

4.7.3 Technical documentation. Technical documentation to be used in the demonstration shall be limited to the technical manual.

4.7.4 Rejection. Failure to conform to the specified requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the Command or Agency concerned shall be immediately notified. The contractor shall, at no additional cost to the Government:

(a) Develop an approach for redesign or correction of all deficiencies.
(b) Upon approval of a documented approach, the contractor shall repeat the demonstration until an accept decision is reached.

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4.8 Inspection of preparation for delivery. Inspections shall be conducted to ensure compliance with Section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Packing. Each switchboard shall be packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity.

5.2 Unitized loads. When applicable, items packed as specified in 5.1 shall be palletized or containerized in accordance with the requirements of MIL-STD-147 whenever quantities to one destination equal 1.1329 cubic meters (40 cubic feet) or more. Quantities of packed items too small or impracticable for a unitized load shall be consolidated in shipping container as specified in 5.1. Palletized loads shall be strapped to the pallet. When applicable, items for shipboard end destinations shall be unitized by hull number. Unitized loads shall not be subjected to rough handling tests.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and exterior container shall be marked in accordance with the requirements of MIL-STD-129. Marking shall include special marking for Method II packs.

5.4 General. Exterior containers shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent possible. Unless otherwise specified, loose-fill polystyrene shall not be used for packaging or packing. Adhesives shall not be used in the closure of wraps, bags, or boxes.

6. NOTES

6.1 Intended use. This equipment is intended to be a one way replacement switchboard for the SB-2727/SRR. Unit will be used in conjunction with the following equipment:

(a) R-389/URR
(b) R-390/URR
(c) AM-215/U
(d) VP-107-4
(e) C-1138B/U
(f) AN/UCC-1
(g) AN/SGC-1
(h) Audio terminal equipments (for example, head sets)

6.2 Ordering data. The procurement document should specify the following:

(a) Title, number, and date of this specification
(b) Number of first article samples to be submitted if other than specified in 4.3
(c) Levels of preservation-packaging, packing, and applicable markings

Preparing activity:
Navy-EC
(Project 5820-N666)

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SPECIFICATION ANALYSIS SHEET

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Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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